

Finish the following tasks, save the output tables (.csv) and figures in the report (.pdf). Submit the zip file.

1. In R workspace, load the dataset by:

```
rawCounts <- read.delim("http://genomedata.org/gen-viz-workshop/intro_to_deseq2/tutorial/E-GEOD-50760-raw-counts.tsv")
head(rawCounts)
```

```
# Read in the sample mappings
sampleData <- read.delim("http://genomedata.org/gen-viz-workshop/intro_to_deseq2/tutorial/E-GEOD-50760-experiment-design.tsv")
head(sampleData)
```

2. Read the manual in <https://bioconductor.org/packages/release/bioc/html/DESeq2.html>, then run DESeq2 to identify differential expressed (DE) genes between tumor and normal. List the DE genes by p-value (save as .csv file).
3. Conduct gene set enrichment to the identified DE genes. List the function that domain the difference between tumor and normal. Describe your conclusion with the table and figures generated by gene set enrichment.
4. Apply hierarchical clustering to genes, then draw the cluster dendrogram and heatmap. (Ref https://tavareshugo.github.io/data-carpentry-rnaseq/04b_rnaseq_clustering.html)
5. (Optional) Conduct gene set enrichment to each gene cluster.